

# Swayam Singh

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## EDUCATION

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### University of Allahabad

2020 – 2024

B.Tech, Computer Science and Engineering

CGPA: 9.56

## EXPERIENCE

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### Microsoft Research

10/2024 – Present

Research Fellow

- Developed & trained the world's first model and training pipeline for End-To-End Autoformalization and verification of programs in Rust
- Engineered the mid-training and post-training recipe for model learn formalizing program with formal guarantee, translating natural-language and informal specs into formally verifiable representations.
- Developing foundational algorithms that exploit reinforcement-learning gradient-optimization dynamics to improve representation efficiency, stability, and generalization.
- Developed an LLM on par with Gemini and GPT-4o for code generation, instruction-following, and reasoning; post-pretraining and online/offline RL.
- Developed SeleKT, a supervised fine-tuning method that preserves generalization while significantly improving downstream performance.
- Led large-scale training on multi-GPU clusters, optimizing distributed workflows for high-performance code synthesis.

### NumPy (numpy-quaddtype)

12/2024 – Present

Maintainer

- Review, merge, and evaluate core PRs affecting dtype architecture, numerical semantics, and cross-platform behavior.
- Maintain core NumPy with a focus on dtype infrastructure, numerical correctness, and long-term maintenance.
- Work with contributors across architectures on portability, performance, and ABI concerns.

### Quansight

07/2024 – 09/2024

Open Source Intern – Numerical Types & Precision (NumPy OSS)

- Led development of NumPy-QuadDType, a cross-platform 128-bit floating-point system for consistent quad precision beyond platform-dependent long double.
- Implemented full dtype ecosystem integration: casting, ufunc dispatch, scalar types, memory model, and serialization.
- Built multi-platform distribution and testing infrastructure handling compiler toolchains, SIMD availability, FMA support, and architecture-specific behavior.

### dataX.ai

05/2022 – 10/2022

Machine Learning Engineer Intern

- Built deep-learning models for vision and language; created an ONNX conversion API with Triton deployment, cutting VM load by 12%.
- Wrote custom CUDA kernels for 3D medical-scan processing, achieving a 2x speedup in segmentation over existing GPU solutions.

### Scaler (InterviewBit)

2022

Data Science Intern

- Built predictive models and data-preprocessing automation, improving user engagement by ~25%.

## PUBLICATIONS

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- **NextCoder: Robust Adaptation of Code LMs to Diverse Code Edits** (ICML 2025; ICLR 2025 DL4Code). **Swayam Singh**, Tushar Aggarwal, et al. A synthetic-data generation pipeline and SeleKT (Selective Knowledge Transfer), a new fine-tuning algorithm that makes code LLMs robust to diverse, real-world code edits. Our 32B model matches Gemini and beats GPT-4o on harder tasks like aider-polyglot, achieved purely via SFT, with SeleKT substantially reducing catastrophic forgetting compared to other methods.
- **Narrow Transformer: StarCoder-Based Java-LM For Desktop** (arXiv:2407.03941, 2024). Kamalkumar Rathinasamy, ..., **Swayam Singh**, et al. A compact, Java-specialized code language model designed to run efficiently on desktop hardware.

- **OctoPack: Instruction Tuning Code Large Language Models** (ICLR 2024 Spotlight; NeurIPS 2023 Instruction Workshop). Niklas Muennighoff, . . . , **Swayam Singh**, et al. Instruction tuning of code models using natural-language Git commits (CommitPack / CommitPackFT).
- **StarCoder: May the Source Be With You!** (TMLR 2023). Raymond Li, . . . , **Swayam Singh**, et al. (BigCode). A 15.5B-parameter open code LLM trained on 1T tokens of permissively licensed code.

## PROJECTS

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- **Virtual Clothing Assistant (PyTorch)**: An end-to-end virtual try-on system: ResNet101 and UNet for garment/body segmentation, OpenPose for pose estimation, and a PyTorch VITON pipeline for realistic garment warping. Reaches an SSIM of 0.895 and has 500+ GitHub stars.
- **Numpy-QuadDType (C, C++, Python)**: A cross-platform 128-bit (quadruple-precision) floating-point dtype for NumPy with 100k+ downloads. A portable alternative to long double with full casting, ufunc dispatch, scalar types, and serialization that behaves consistently across compilers and architectures.
- **Bare-Bones Inference (CUDA, C++)**: A minimal inference stack with engine-grade features such as batching and scheduling, but serving a single fused megakernel instead of a conventional multi-kernel pipeline. Built DeepSeek-V2 (236B)’s block-scaled FP8-quantized, multi-GPU megakernel for NVIDIA B200 (sm\_100) GPUs that fuses all computation and TP/EP communication into one kernel, handling cross-rank exchange via `multimem` PTX multicast writes (directly writing into peer ranks’ memory at once). Achieves 13.11 ms/token decode latency, beating vLLM’s optimized serving latency at  $B=1$  by 2x and eager-mode by >10x.
- **QBLAS (C, C++)**: A high-performance BLAS library for IEEE-754 binary128 (quadruple) precision, implementing optimized linear-algebra kernels that bring quad-precision numerics to workloads unsupported by standard double-precision libraries.
- **cpp-verify (C++, LLVM, SMT)**: Extends C++ with first-class formal-verification constructs (pre/post-conditions and invariants), lowering specifications through an LLVM-based pipeline and discharging the resulting proof obligations to SMT solvers.
- **MIRA – Multimodal Image Reconstruction with Attention (PyTorch)**: A transformer-based architecture for single-view text/image-to-3D reconstruction using ViT encoders and triplane decoders with cross-attention. Generates 3D mesh and video in under 10s on an A100, with SDXL-integrated diffusion for end-to-end scene synthesis.

## HONORS AND AWARDS

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- **2024 — Kaggle Competition Expert**: Bronze medal (top 7%) in the UBC-OCEAN competition; top 3% in the *30 Days of ML* challenge.
- **2024 — Invited to Google Research Week**, Google Research’s gathering of AI researchers (keynote by Jeff Dean; sessions on differential privacy, responsible AI, and more).
- **2024 — OctoPack** accepted as a **Spotlight (top 5%)** at ICLR 2024.
- **2023 — Selected for the Amazon ML Summer School 2023.**
- **2023 — Clothes Virtual Try-On** crossed 500+ GitHub stars.

## INVITED TALKS

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- **MAMBA: Zero to Hero**, invited talk on State Space Models at Cohere for AI.
- **Provably-Correct Code and Efficient Sparse Training of LLMs**, internal research talk at Microsoft Research.
- **Foundations of Machine Learning**, a GDG On-Campus session on the ML landscape, tooling ecosystem, and emerging directions.
- **From Deep Learning to Large Language Models**, a talk on the foundations of modern AI: deep learning, generative models, and LLMs.

## SKILLS SUMMARY

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- **Languages**: C, C++, CUDA, CuTeDSL, Python
- **ML & Training**: LLM post-training, reinforcement learning (online/offline RL), supervised fine-tuning, multi-GPU distributed training, code-generation models
- **Systems & GPU**: CUDA kernels and megakernels, high-performance numerical computing, IEEE-754 quad precision, SIMD/FMA, performance optimization
- **Tools & Frameworks**: PyTorch, NumPy (C-API internals), LLVM, SMT solvers
- **Open Source**: Core NumPy maintenance, cross-platform C-extension packaging, code review